ObdTuner information

ObdTuner is the perfect starting point for tuning the Z22SE (L61) engine. ObdTuner is used in over 15 different countries and with engines having received a broad range of modifications. During this time ObdTuner has proven to be very reliable and easy to use, especially for people who are not an ECU tuning expert.

The free accompanying PC software has all the necessary features to tune the engine and also has lots of features for easy problem solving. ObdTuner will make the car feel like the modifications done to the engine are factory installed so no start or warming up problems, clean exhaust gases, high fuel mileage and optimum power delivery under all conditions.

The software can be used with all kind of engine modifications and can easily be adapted to a new engine set-up without extra software costs. It is not necessary to start from scratch, because of the large user base there are lots of calibrations available which can be used as a starting point. For ObdTuner users, these base calibrations are free. When the ECU is updated to ObdTuner a base calibration will be installed which is adapted to the (new) engine set-up so the ECU will work with your engine out of the box.

ObdTuner contains all necessary features for adapting your ECU to the modifications done to your engine. ObdTuner stands at the base of many projects and made it possible to explore the full potential of the Z22SE (L61) engine. Lately ObdTuner is also used with other members from the Ecotec family which are often used as a replacement engine for the Z22SE (L61) because of their similar dimensions.

With ObdTuner all OEM safety and error checks including knock control etc. are still fully operational. Exhaust emissions will stay road legal (for this a catalytic converter has to be installed). Also a large part or even the complete original wiring loom can be re-used so the original day to day reliability is preserved.

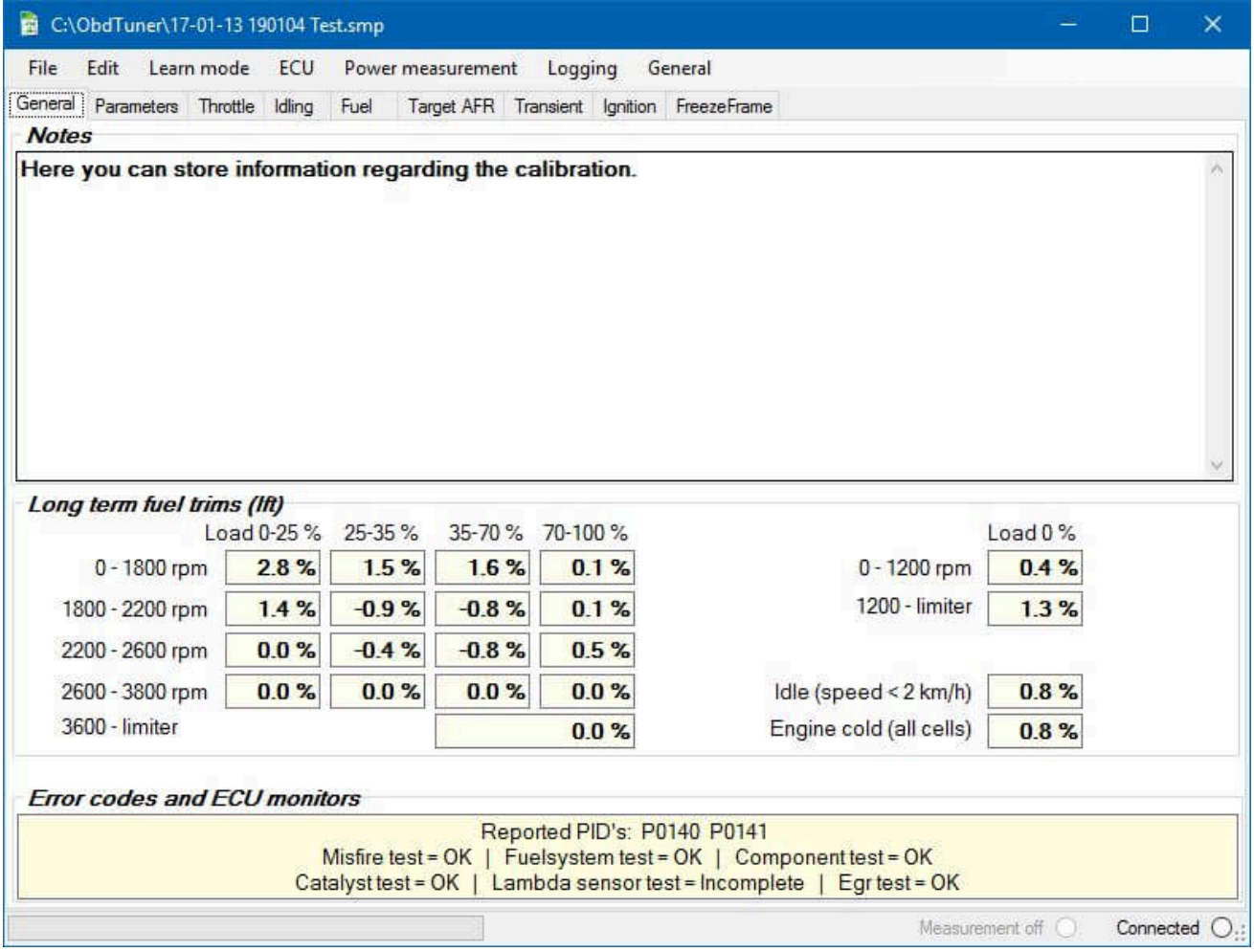
ObdTuner is very often used as the software of choice for engines receiving an Eaton M62 supercharger. The Eaton M62 supercharger is an OEM bold on supercharger for the Ecotec family and is used on the GM LSJ engine. The Z22SE (L61) is strong enough to withstand the added extra torque and power without the need to replace internal parts, this modification is even promoted by GM itself. For more information have a look at the 'GM Sport Compact Performance Build Book' part 'Engine Assembly Stock To 250 HP'.

Some screen shots

All values in the screen shots below, except the general parameters, are live programmable. This means, when ObdTuner is connected to the car, changing a value in one of the tables will also immediately change the corresponding value in the ECU. This can be very handy, for instance; changing the throttle response can be done while driving, and you will get instant feedback.

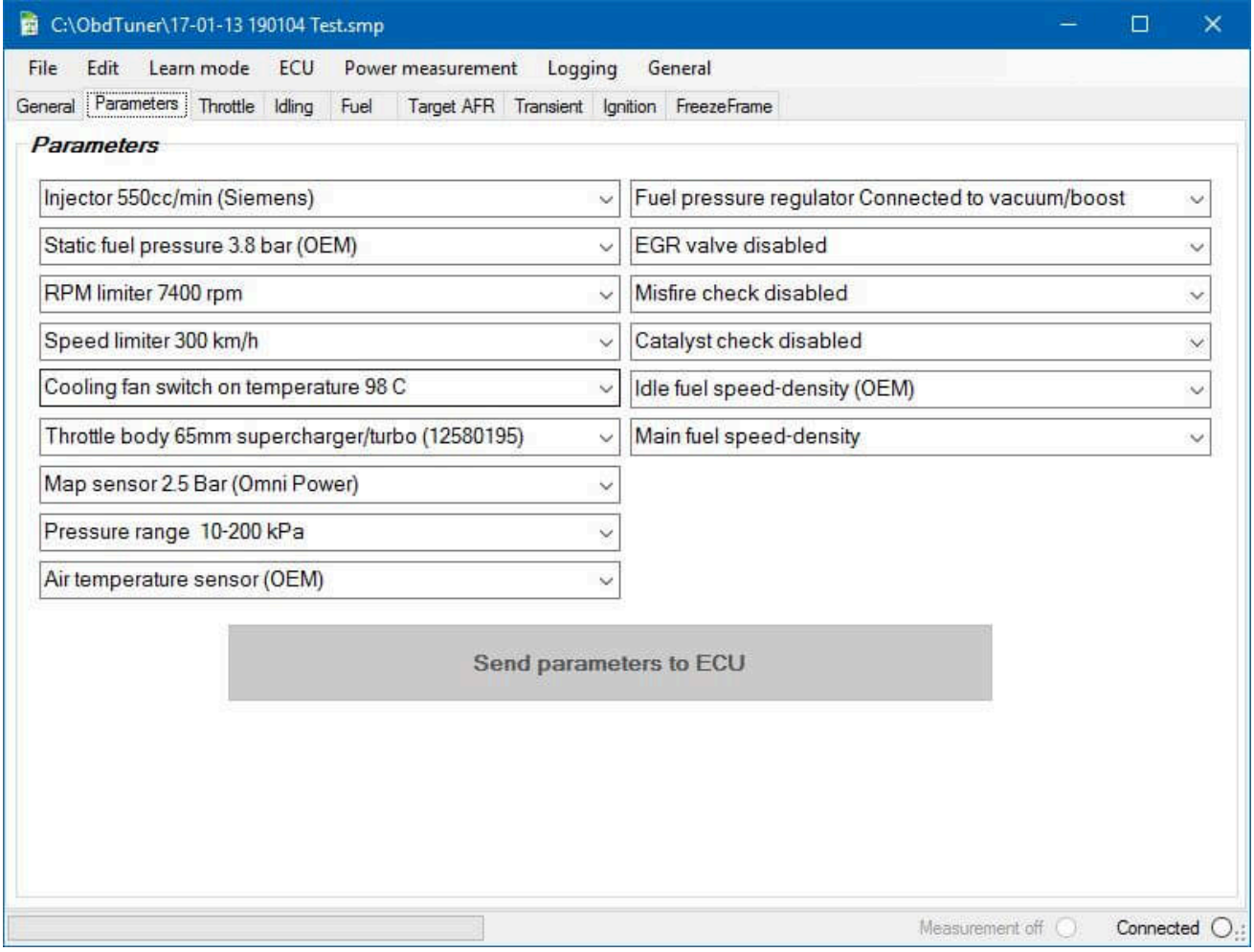
General tab

This is the start-up screen, here you can enter session information, take a look at the long-term fuel trims and see if there are error codes set.



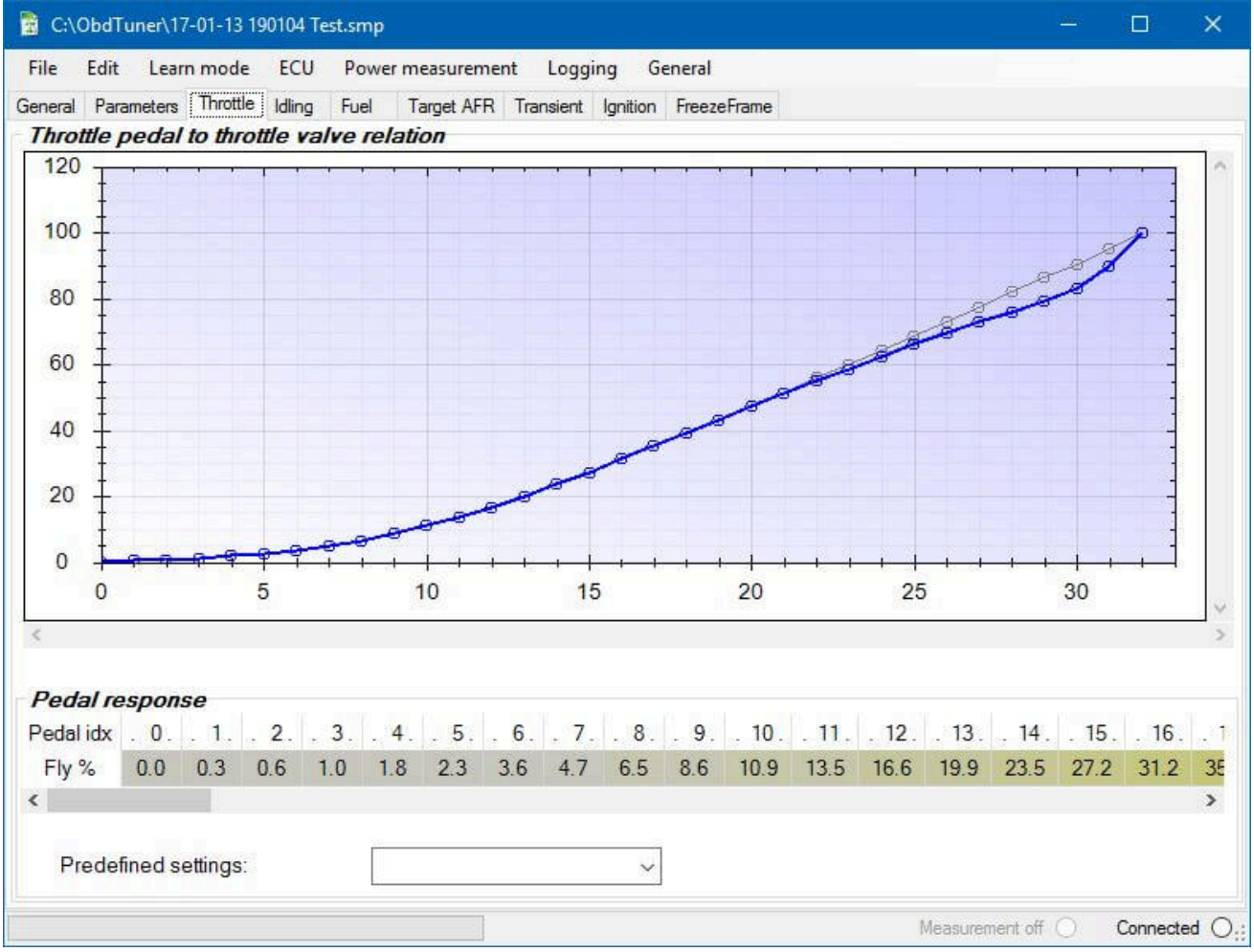
Parameters tab

Here engine set up parameters can be configured like the installed injector type etc.



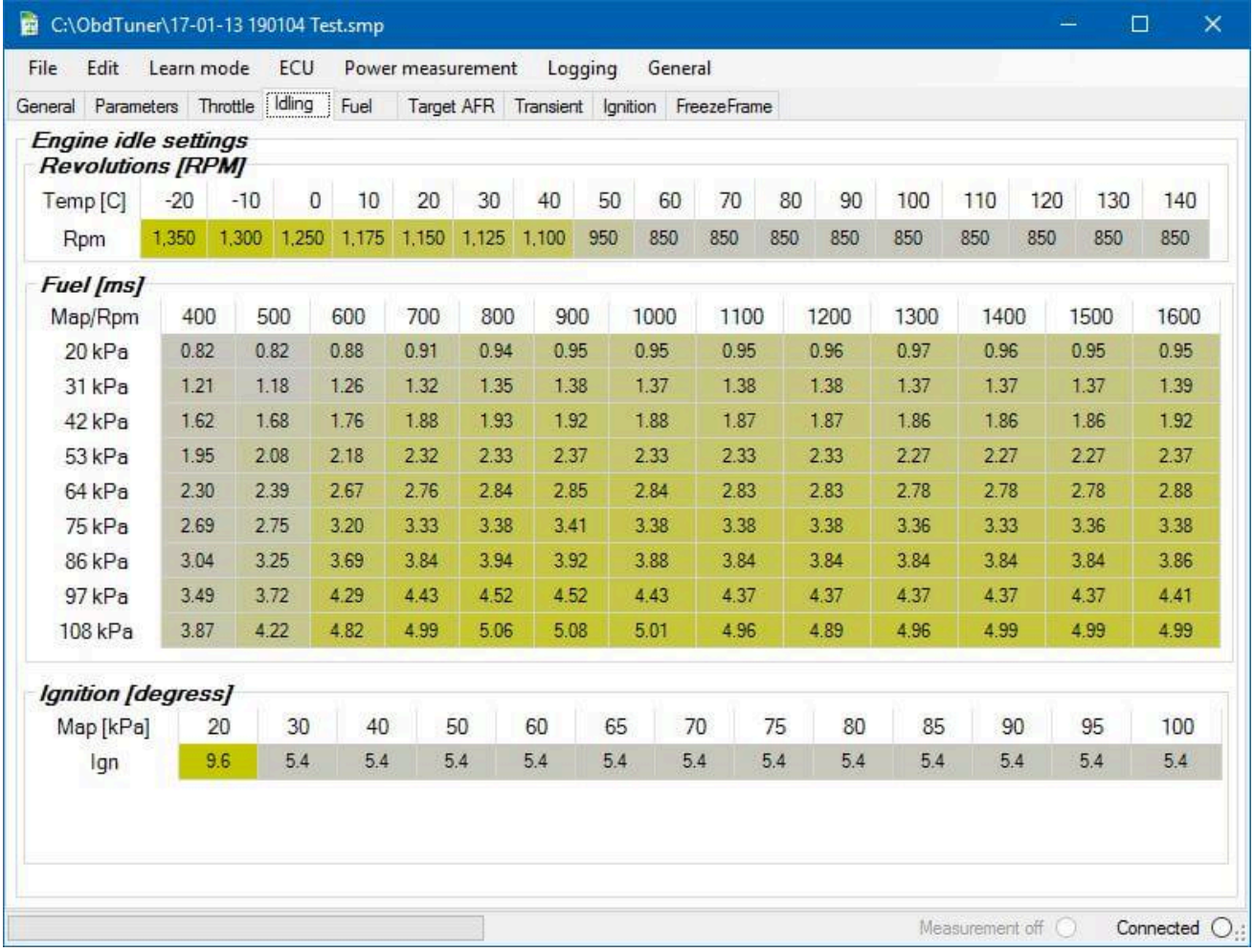
Throttle tab

Here you can configure the throttle pedal position versus the throttle fly position.



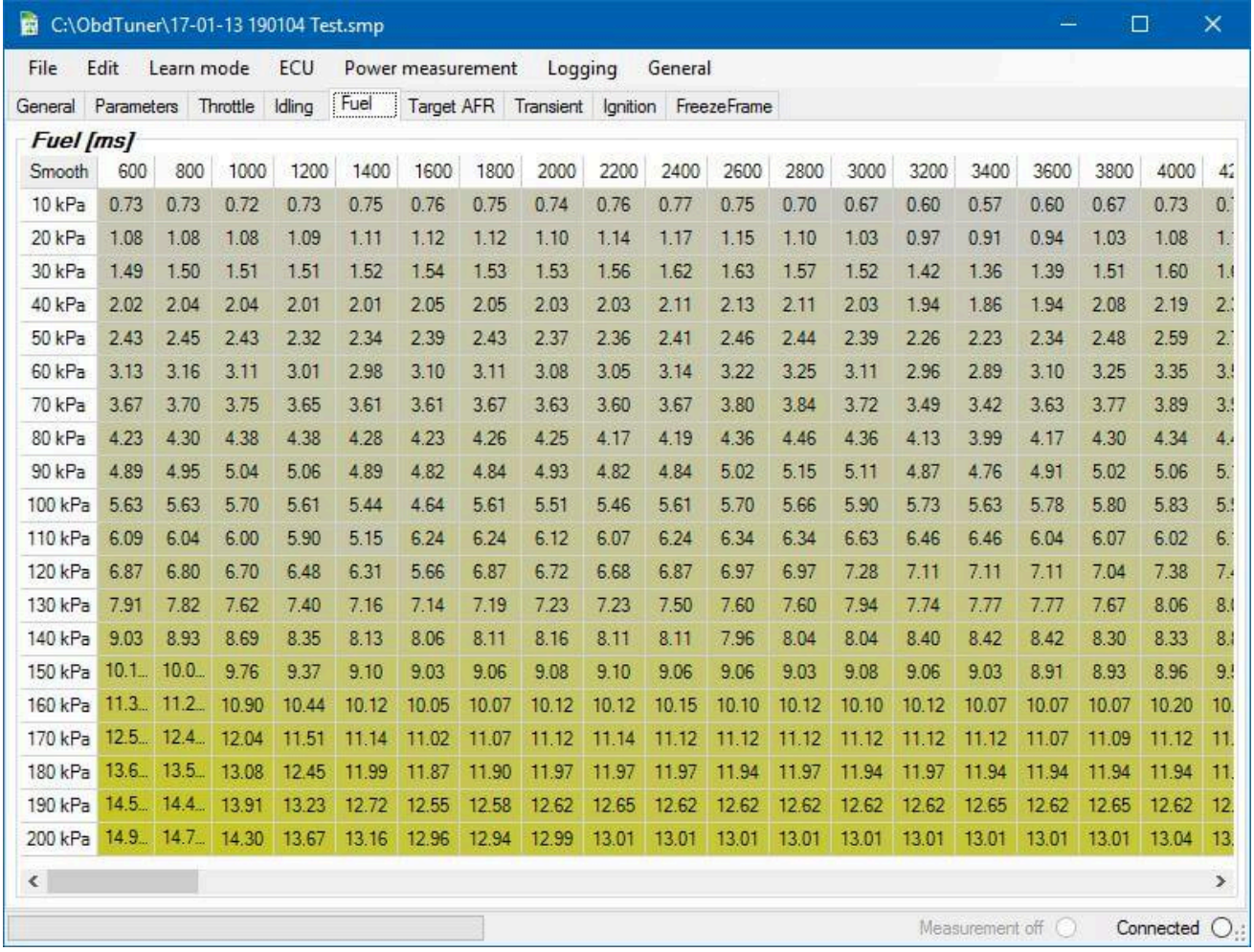
Idling tab

Here the idling settings like idling rpm, ignition advance and fuel quantity can be configured.



Fuel tab

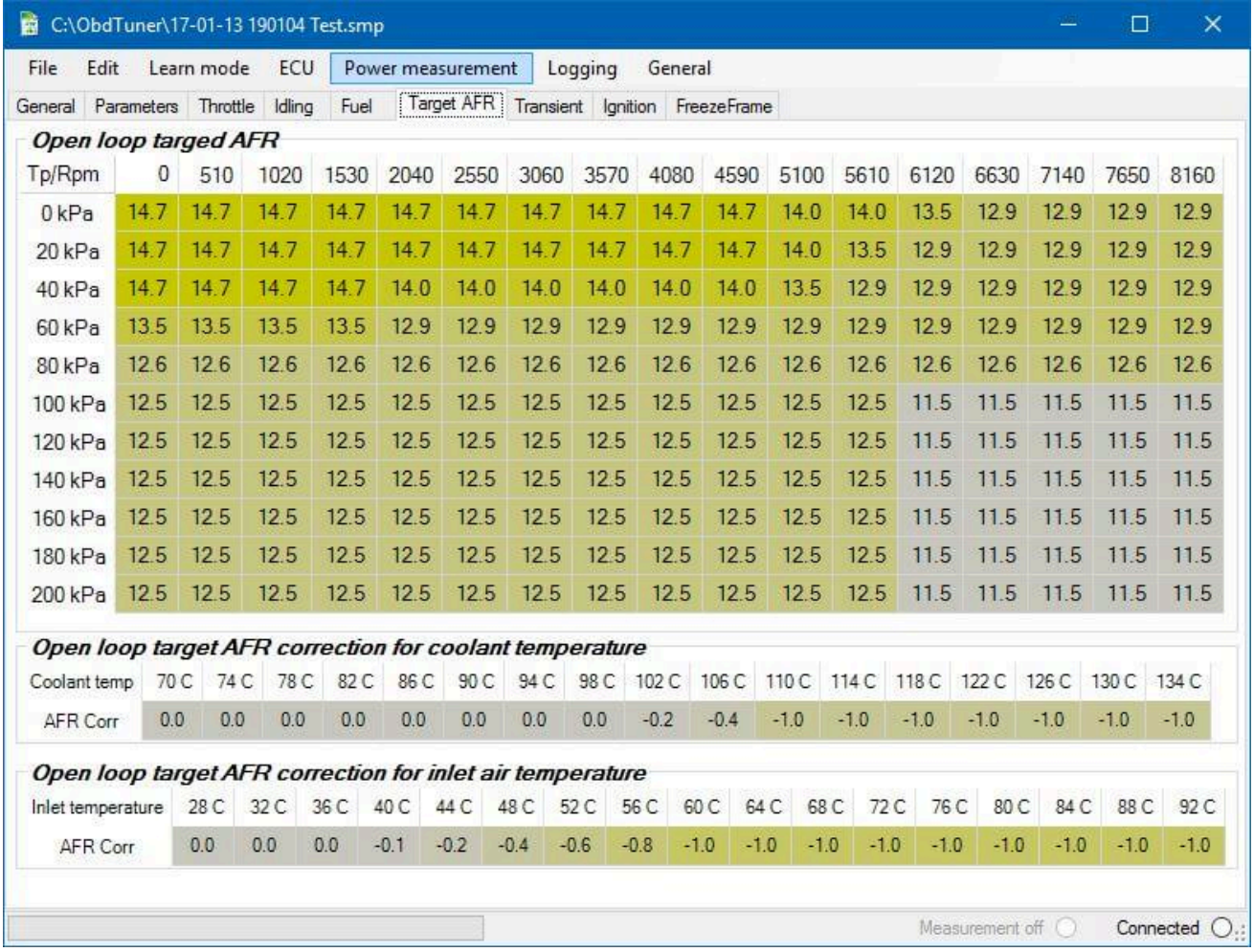
The fuel tab is for configuring the main fuel quantity.



Smooth	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800	3000	3200	3400	3600	3800	4000	4
10 kPa	0.73	0.73	0.72	0.73	0.75	0.76	0.75	0.74	0.76	0.77	0.75	0.70	0.67	0.60	0.57	0.60	0.67	0.73	0.
20 kPa	1.08	1.08	1.08	1.09	1.11	1.12	1.12	1.10	1.14	1.17	1.15	1.10	1.03	0.97	0.91	0.94	1.03	1.08	1.
30 kPa	1.49	1.50	1.51	1.51	1.52	1.54	1.53	1.53	1.56	1.62	1.63	1.57	1.52	1.42	1.36	1.39	1.51	1.60	1.
40 kPa	2.02	2.04	2.04	2.01	2.01	2.05	2.05	2.03	2.03	2.11	2.13	2.11	2.03	1.94	1.86	1.94	2.08	2.19	2.
50 kPa	2.43	2.45	2.43	2.32	2.34	2.39	2.43	2.37	2.36	2.41	2.46	2.44	2.39	2.26	2.23	2.34	2.48	2.59	2.
60 kPa	3.13	3.16	3.11	3.01	2.98	3.10	3.11	3.08	3.05	3.14	3.22	3.25	3.11	2.96	2.89	3.10	3.25	3.35	3.
70 kPa	3.67	3.70	3.75	3.65	3.61	3.61	3.67	3.63	3.60	3.67	3.80	3.84	3.72	3.49	3.42	3.63	3.77	3.89	3.
80 kPa	4.23	4.30	4.38	4.38	4.28	4.23	4.26	4.25	4.17	4.19	4.36	4.46	4.36	4.13	3.99	4.17	4.30	4.34	4.
90 kPa	4.89	4.95	5.04	5.06	4.89	4.82	4.84	4.93	4.82	4.84	5.02	5.15	5.11	4.87	4.76	4.91	5.02	5.06	5.
100 kPa	5.63	5.63	5.70	5.61	5.44	4.64	5.61	5.51	5.46	5.61	5.70	5.66	5.90	5.73	5.63	5.78	5.80	5.83	5.
110 kPa	6.09	6.04	6.00	5.90	5.15	6.24	6.24	6.12	6.07	6.24	6.34	6.34	6.63	6.46	6.46	6.04	6.07	6.02	6.
120 kPa	6.87	6.80	6.70	6.48	6.31	5.66	6.87	6.72	6.68	6.87	6.97	6.97	7.28	7.11	7.11	7.11	7.04	7.38	7.
130 kPa	7.91	7.82	7.62	7.40	7.16	7.14	7.19	7.23	7.23	7.50	7.60	7.60	7.94	7.74	7.77	7.77	7.67	8.06	8.
140 kPa	9.03	8.93	8.69	8.35	8.13	8.06	8.11	8.16	8.11	8.11	7.96	8.04	8.04	8.40	8.42	8.42	8.30	8.33	8.
150 kPa	10.1...	10.0...	9.76	9.37	9.10	9.03	9.06	9.08	9.10	9.06	9.06	9.03	9.08	9.06	9.03	8.91	8.93	8.96	9.
160 kPa	11.3...	11.2...	10.90	10.44	10.12	10.05	10.07	10.12	10.12	10.15	10.10	10.12	10.10	10.12	10.07	10.07	10.07	10.20	10.
170 kPa	12.5...	12.4...	12.04	11.51	11.14	11.02	11.07	11.12	11.14	11.12	11.12	11.12	11.12	11.12	11.12	11.07	11.09	11.12	11.
180 kPa	13.6...	13.5...	13.08	12.45	11.99	11.87	11.90	11.97	11.97	11.97	11.94	11.97	11.94	11.97	11.94	11.94	11.94	11.94	11.
190 kPa	14.5...	14.4...	13.91	13.23	12.72	12.55	12.58	12.62	12.65	12.62	12.62	12.62	12.62	12.62	12.65	12.62	12.65	12.62	12.
200 kPa	14.9...	14.7...	14.30	13.67	13.16	12.96	12.94	12.99	13.01	13.01	13.01	13.01	13.01	13.01	13.01	13.01	13.01	13.04	13.

Target AFR tab

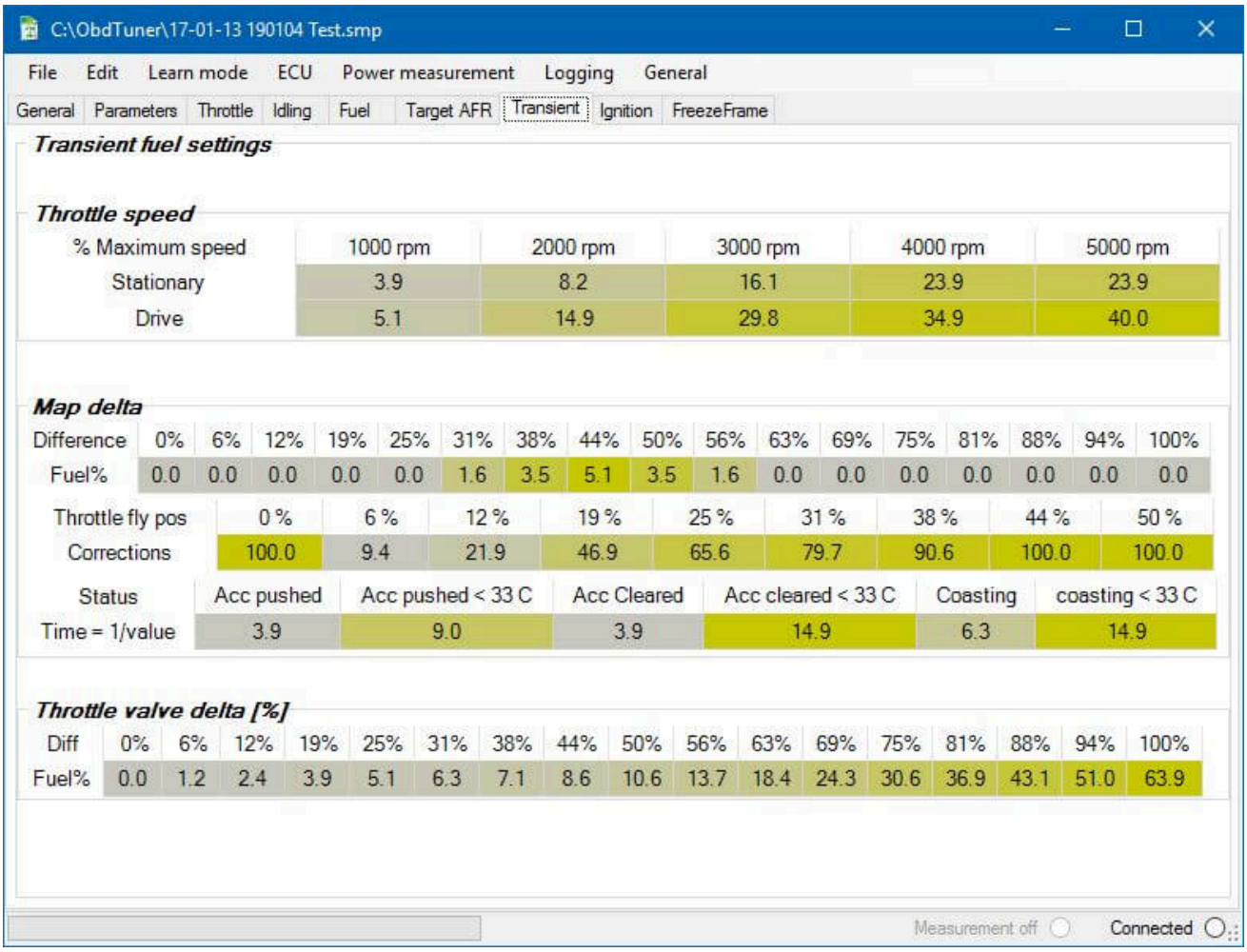
Here the target AFR can be configured, this is the Air To Fuel ratio the ECU will use when going to open loop (full throttle).



Transient fuel tab

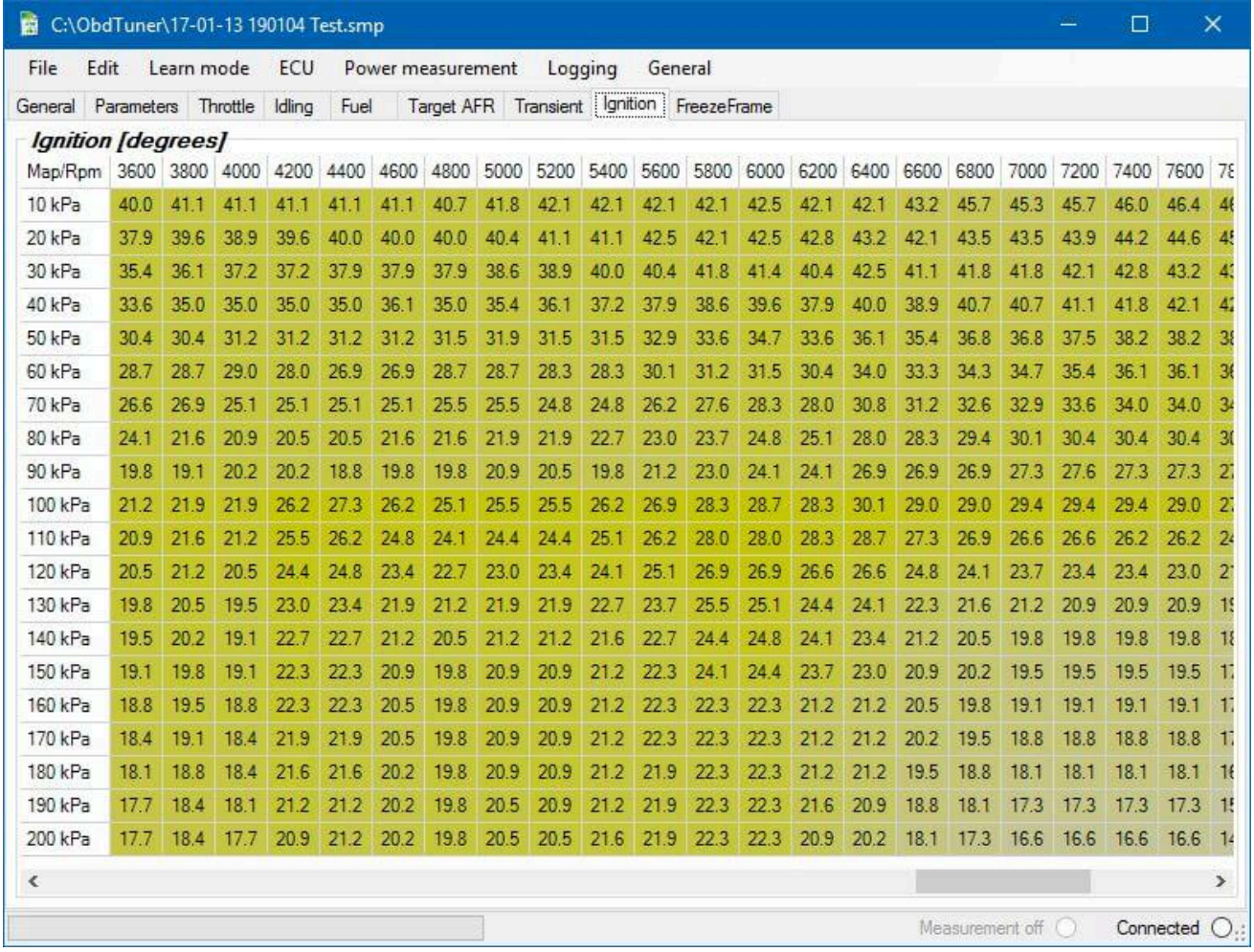
If the engine is not responding instantly when pushing the throttle pedal it most often is because the fuelling is not spot on in transient conditions, this tab is the place to optimize

this.



Ignition tab

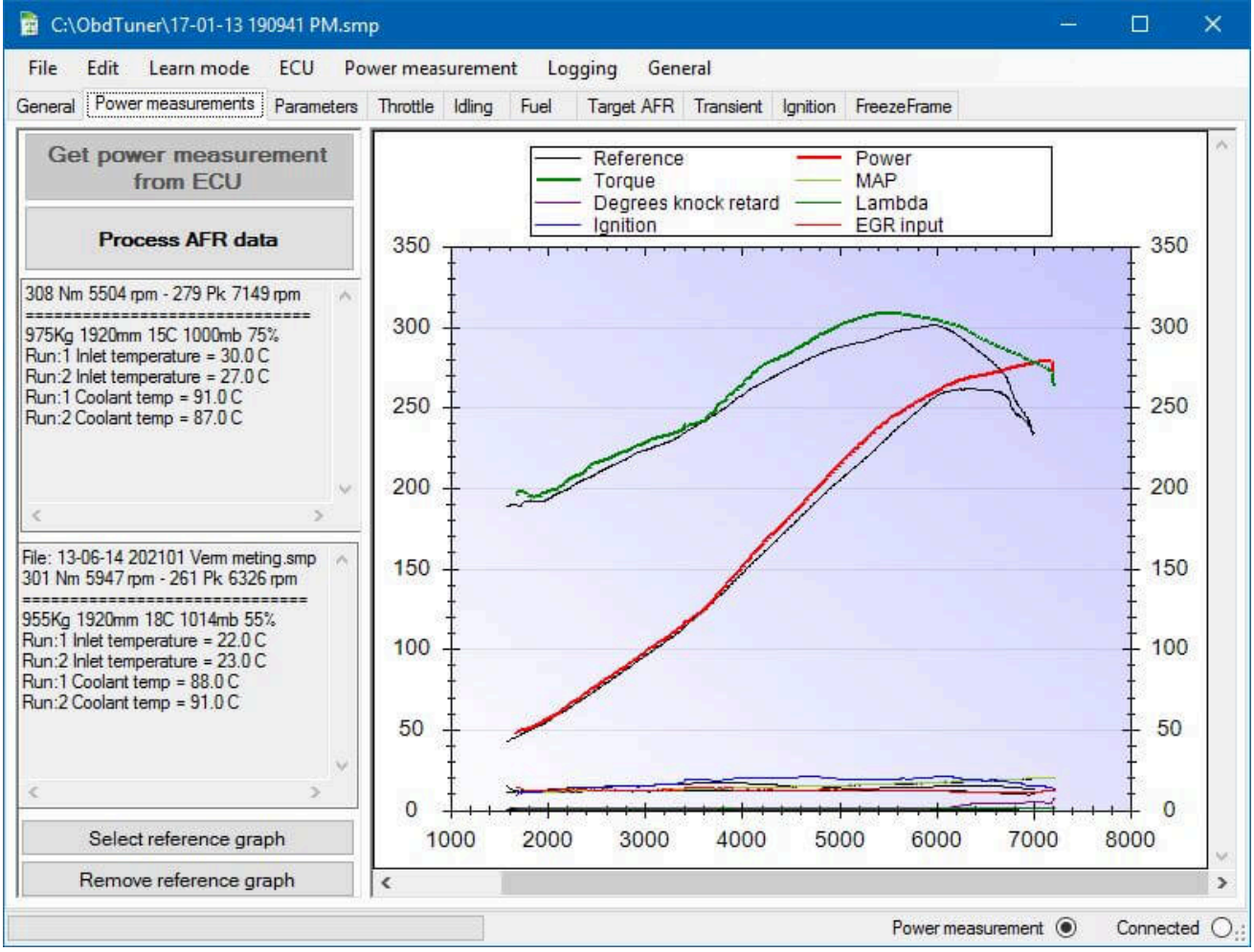
This is the main ignition table, here the ignition can be adapted to the engine set-up and fuel.



Map/Rpm	3600	3800	4000	4200	4400	4600	4800	5000	5200	5400	5600	5800	6000	6200	6400	6600	6800	7000	7200	7400	7600	78
10 kPa	40.0	41.1	41.1	41.1	41.1	41.1	40.7	41.8	42.1	42.1	42.1	42.1	42.5	42.1	42.1	43.2	45.7	45.3	45.7	46.0	46.4	46
20 kPa	37.9	39.6	38.9	39.6	40.0	40.0	40.0	40.4	41.1	41.1	42.5	42.1	42.5	42.8	43.2	42.1	43.5	43.5	43.9	44.2	44.6	45
30 kPa	35.4	36.1	37.2	37.2	37.9	37.9	37.9	38.6	38.9	40.0	40.4	41.8	41.4	40.4	42.5	41.1	41.8	41.8	42.1	42.8	43.2	43
40 kPa	33.6	35.0	35.0	35.0	35.0	36.1	35.0	35.4	36.1	37.2	37.9	38.6	39.6	37.9	40.0	38.9	40.7	40.7	41.1	41.8	42.1	42
50 kPa	30.4	30.4	31.2	31.2	31.2	31.2	31.5	31.9	31.5	31.5	32.9	33.6	34.7	33.6	36.1	35.4	36.8	36.8	37.5	38.2	38.2	38
60 kPa	28.7	28.7	29.0	28.0	26.9	26.9	28.7	28.7	28.3	28.3	30.1	31.2	31.5	30.4	34.0	33.3	34.3	34.7	35.4	36.1	36.1	36
70 kPa	26.6	26.9	25.1	25.1	25.1	25.1	25.5	25.5	24.8	24.8	26.2	27.6	28.3	28.0	30.8	31.2	32.6	32.9	33.6	34.0	34.0	34
80 kPa	24.1	21.6	20.9	20.5	20.5	21.6	21.6	21.9	21.9	22.7	23.0	23.7	24.8	25.1	28.0	28.3	29.4	30.1	30.4	30.4	30.4	30
90 kPa	19.8	19.1	20.2	20.2	18.8	19.8	19.8	20.9	20.5	19.8	21.2	23.0	24.1	24.1	26.9	26.9	26.9	27.3	27.6	27.3	27.3	27
100 kPa	21.2	21.9	21.9	26.2	27.3	26.2	25.1	25.5	25.5	26.2	26.9	28.3	28.7	28.3	30.1	29.0	29.0	29.4	29.4	29.4	29.0	27
110 kPa	20.9	21.6	21.2	25.5	26.2	24.8	24.1	24.4	24.4	25.1	26.2	28.0	28.0	28.3	28.7	27.3	26.9	26.6	26.6	26.2	26.2	24
120 kPa	20.5	21.2	20.5	24.4	24.8	23.4	22.7	23.0	23.4	24.1	25.1	26.9	26.9	26.6	26.6	24.8	24.1	23.7	23.4	23.4	23.0	27
130 kPa	19.8	20.5	19.5	23.0	23.4	21.9	21.2	21.9	21.9	22.7	23.7	25.5	25.1	24.4	24.1	22.3	21.6	21.2	20.9	20.9	20.9	19
140 kPa	19.5	20.2	19.1	22.7	22.7	21.2	20.5	21.2	21.2	21.6	22.7	24.4	24.8	24.1	23.4	21.2	20.5	19.8	19.8	19.8	19.8	18
150 kPa	19.1	19.8	19.1	22.3	22.3	20.9	19.8	20.9	20.9	21.2	22.3	24.1	24.4	23.7	23.0	20.9	20.2	19.5	19.5	19.5	19.5	17
160 kPa	18.8	19.5	18.8	22.3	22.3	20.5	19.8	20.9	20.9	21.2	22.3	22.3	22.3	21.2	21.2	20.5	19.8	19.1	19.1	19.1	19.1	17
170 kPa	18.4	19.1	18.4	21.9	21.9	20.5	19.8	20.9	20.9	21.2	22.3	22.3	22.3	21.2	21.2	20.2	19.5	18.8	18.8	18.8	18.8	17
180 kPa	18.1	18.8	18.4	21.6	21.6	20.2	19.8	20.9	20.9	21.2	21.9	22.3	22.3	21.2	21.2	19.5	18.8	18.1	18.1	18.1	18.1	16
190 kPa	17.7	18.4	18.1	21.2	21.2	20.2	19.8	20.5	20.9	21.2	21.9	22.3	22.3	21.6	20.9	18.8	18.1	17.3	17.3	17.3	17.3	15
200 kPa	17.7	18.4	17.7	20.9	21.2	20.2	19.8	20.5	20.5	21.6	21.9	22.3	22.3	20.9	20.2	18.1	17.3	16.6	16.6	16.6	16.6	14

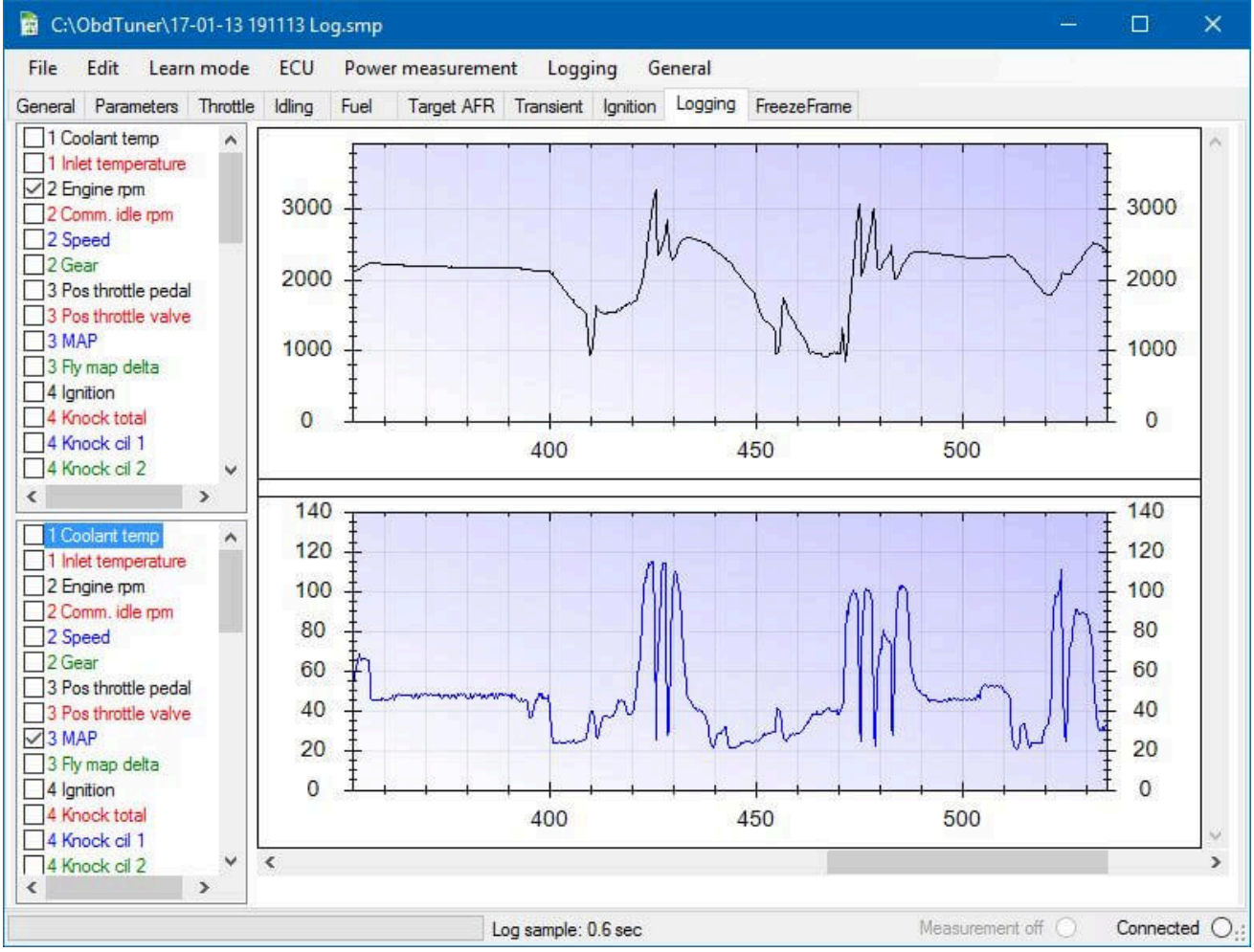
Power measurement tab

The ECU can be set in power measurement mode, this way it is very easy to make a power run. When the weather data, vehicle weight, etc is configured correctly it will produce very consistent figures.



Logging tab

This is the data logging tab. All values in the list are logged as a snapshot about 3 times per second. Afterwards or during driving you can select the values you want to monitor and relate them to the values in the other table.



Measurement tab

There is also a fuel learning mode. The ECU will monitor the engine and will compose a correction fuel table which can be automatically integrated in the original fuel table.

C:\ObdTuner\17-01-13 191508 Fuel learning.smp

File Edit Learn mode ECU Power measurement Logging General

General | Measurement | Parameters | Throttle | Idling | Fuel | Target AFR | Transient | Ignition | Logging

Process measurements ☑ Nr of measurements ☑ Fuel correction ☐ Knock retard

Tp/Rpm	1400	1600	1800	2000	2200	2400	2600	2800	3000	3200	3400	360
10 kPa	--	--	--	--	--	--	--	--	--	--	--	
20 kPa	3028 / -5.5	4569 / -3.1	2821 / -6.2	1122 / -5.5	770 / -4.7	1120 / -3.1	522 / -3.9	476 / -4.7	471 / -5.5	384 / -4.7	398 / -3.1	98 /
30 kPa	4372 / -3.9	2778 / -3.9	5451 / -3.1	4237 / -2.3	4293 / -2.3	3176 / -1.6	607 / -2.3	1195 / -2.3	741 / -3.9	872 / -3.1	966 / 0.0	711 /
40 kPa	2710 / -2.3	6977 / -3.1	11595 / -3.1	16581 / -4.7	16411 / -1.6	3480 / -1.6	904 / -2.3	1052 / -1.6	112 / -3.1	284 / -2.3	798 / -2.3	128 /
50 kPa	596 / -2.3	1024 / -3.9	2236 / -1.6	2994 / -2.3	9453 / -1.6	1894 / -1.6	549 / -0.8	1688 / -3.1	260 / -3.1	220 / -2.3	268 / 0.0	324 /
60 kPa	305 / -1.6	930 / -2.3	1494 / -0.8	2335 / -1.6	3200 / -3.9	931 / -2.3	620 / -0.8	956 / -3.9	613 / -2.3	167 / -1.6	483 / -1.6	422 /
70 kPa	177 / -3.1	251 / -3.9	340 / -3.1	680 / -1.6	931 / -3.1	744 / -1.6	466 / -1.6	788 / -2.3	264 / -2.3	245 / -2.3	605 / -1.6	248 /
80 kPa	84 / -3.1	146 / -3.1	174 / -3.9	459 / -3.1	561 / -3.1	378 / -0.8	237 / -0.8	314 / -1.6	238 / -3.1	144 / -0.8	113 / -0.8	47 /
90 kPa	21 / -4.7	27 / -4.7	130 / -4.7	241 / -1.6	418 / -2.3	174 / -3.1	144 / -1.6	235 / -0.8	277 / -3.9	252 / -2.3	229 / 0.0	215 /
100 kPa	3 / -4.7	50 / -5.5	89 / -3.9	125 / -1.6	204 / -3.1	232 / -3.1	345 / -3.1	499 / -1.6	440 / -3.1	305 / -4.7	241 / -3.9	312 /
110 kPa	--	--	9 / -2.3	60 / -1.6	103 / -1.6	191 / -1.6	257 / -2.3	285 / -3.1	335 / -3.1	437 / -3.9	206 / -3.9	82 /
120 kPa	--	--	--	--	--	--	63 / -3.9	133 / -3.1	202 / -3.1	308 / -4.7	306 / -5.5	101 /
130 kPa	--	--	--	--	--	--	--	--	--	10 / -2.3	190 / -3.9	417 /
140 kPa	--	--	--	--	--	--	--	--	--	--	--	
150 kPa	--	--	--	--	--	--	--	--	--	--	--	
160 kPa	--	--	--	--	--	--	--	--	--	--	--	
170 kPa	--	--	--	--	--	--	--	--	--	--	--	
180 kPa	--	--	--	--	--	--	--	--	--	--	--	
190 kPa	--	--	--	--	--	--	--	--	--	--	--	
200 kPa	--	--	--	--	--	--	--	--	--	--	--	

Learning base table ◉ Connected ○